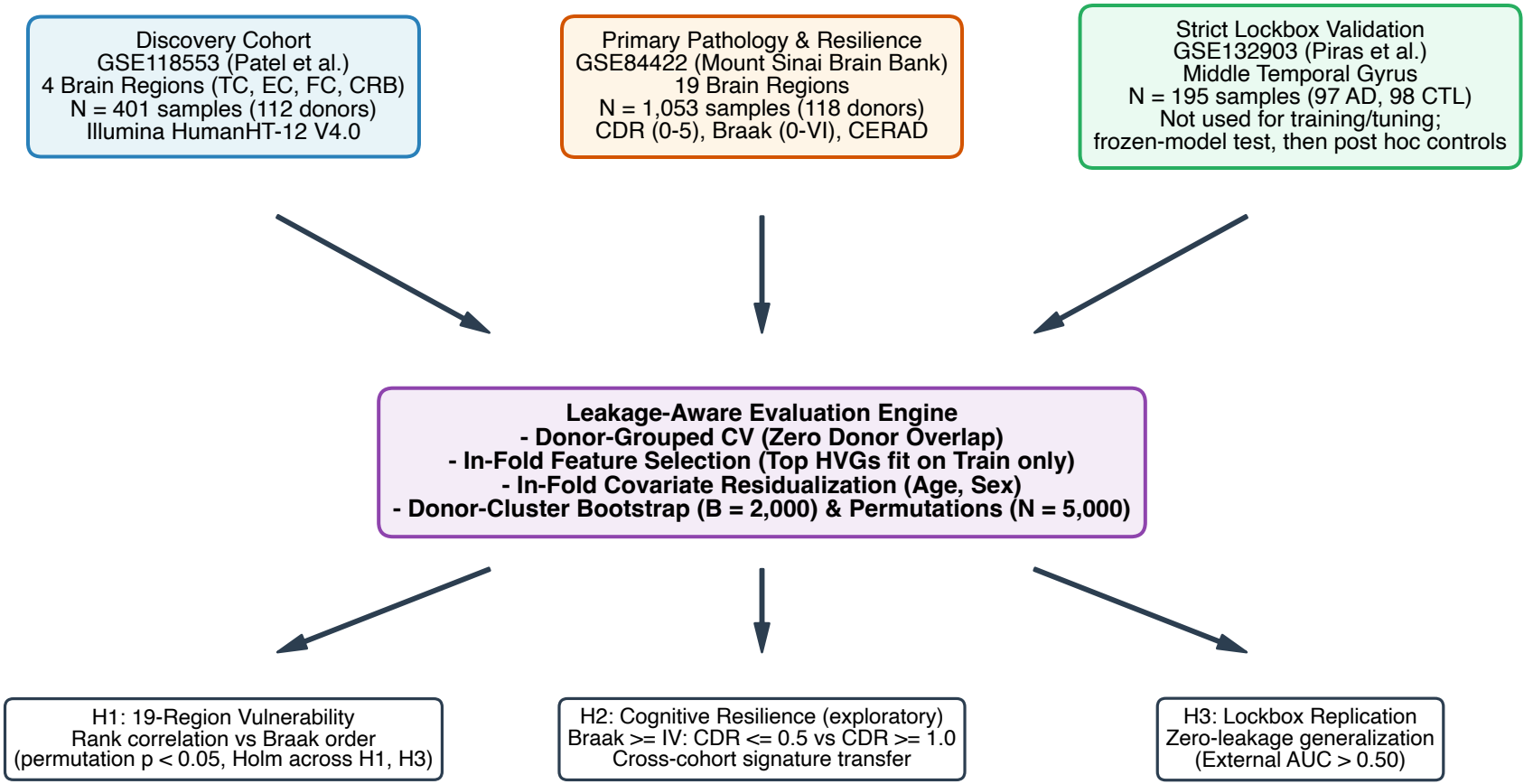
Discovery Cohort
GSE118553 (Patel et al.)
4 Brain Regions (TC, EC, FC, CRB)
N = 401 samples (112 donors)
Illumina HumanHT-12 V4.0
Primary Pathology & Resilience
GSE84422 (Mount Sinai Brain Bank)
19 Brain Regions
N = 1,053 samples (118 donors)
CDR (0-5), Braak (0-VI), CERAD
Strict Lockbox Validation
GSE132903 (Piras et al.)
Middle Temporal Gyrus
N = 195 samples (97 AD, 98 CTL)
Not used for training/tuning;
frozen-model test, then post hoc controls
Leakage-Aware Evaluation Engine
- Donor-Grouped CV (Zero Donor Overlap)
- In-Fold Feature Selection (Top HVGs fit on Train only)
- In-Fold Covariate Residualization (Age, Sex)
- Donor-Cluster Bootstrap (B = 2,000) & Permutations (N = 5,000)
H1: 19-Region Vulnerability
Rank correlation vs Braak order
(permutation p < 0.05, Holm across H1, H3)
H2: Cognitive Resilience (exploratory)
Braak >= IV: CDR <= 0.5 vs CDR >= 1.0
Cross-cohort signature transfer
H3: Lockbox Replication
Zero-leakage generalization
(External AUC > 0.50)